

SEDAR 68 Gulf of Mexico Scamp

Data Workshop

LIFE HISTORY RESEARCH RECOMMENDATIONS

Natural Mortality

- Convene a topical workgroup or other workshop to critically review literature used in Then et al. (2015), discuss recent advancements in ageing approaches (e.g., Gray Triggerfish), and propose best options for selecting species for inclusion in regression analyses for reef fish species in the US Southeast Region to be used in estimating natural mortality.
- Research the Thorson FishLife program for use in natural mortality estimates and measures of uncertainty. <https://github.com/James-Thorson-NOAA/FishLife>

Reproductive Biology

- Investigate the male contribution to spawning success and the potential for sperm limitation in the population through model simulations and field research that will fill in critical gaps in knowledge (i.e., fertilization rate under various sex ratio scenarios, mating strategy) and continue to monitor sex ratio.
- Additional sampling with better spatial and especially temporal coverage to confirm preliminary results that male gonadosomatic index (GSI) indicates that Scamp are spawning in pairs or small groups. This information is lacking for Yellowmouth Grouper.
- Collect all sizes of Yellowmouth Grouper and larger female Scamp (> 650 mm FL) during the spawning season to assess batch fecundity and thereby fill a data gap that prevents estimating total egg production.
- Given the likely smaller population size of Yellowmouth Grouper, samples with a wide range of size/age, from fishery-dependent and fishery-independent sources, are needed to determine reproductive parameters for this species and to allow comparisons with those of Scamp.

COMMERCIAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

- **Recommendation for the use of EM to facilitate the improvement of discard accounting in the South Atlantic**
 - The Center for Electronic Monitoring at Mote (CFEMM) has been applying Electronic Monitoring (EM) in the Gulf of Mexico (GoM) using Saltwater Inc. (SWI) software since 2016.

EM is a valuable monitoring tool for researchers to directly observe and permanently document location, identify bycatch hotspots, catch, effort, and discard data to reduce uncertainty in critical finfish and shark fishery data for use by industry and management.

- In the absence of a robust reef fish observer program in the South Atlantic, the commercial workgroup recognizes EM as a tool to improve discard accounting in the region.

Additionally, the COVID-19 pandemic has hampered interactions between the fishing industry and state/federal fisheries data collections. The workgroup recognizes the potential for work pioneered by the CFEMM to advance biological sampling needs without human observers.

- Continue to explore additional methods, such as citizen science (e.g. SAFMC Scamp Release), to help supplement information to characterize discard size composition

- **Recommendation for South Atlantic and Gulf of Mexico unified methodology in preparation of commercial landings**

- The SEDAR 68 commercial workgroup has recognized that there are significant differences in the South Atlantic and Gulf of Mexico in the approach to the preparation of commercial landings. These differences were identified specifically in discussions of proportioning, validation, and data provision formats.
- In order to resolve the issue, the workgroup recommends that SEDAR staff convene and facilitate a joint-regional workshop for commercial workgroup members from both regions in order to follow-up on and confirm the best practices in Procedural Workshop 7.
- Previous workgroup leaders should be consulted in establishing the TORs for the workshop.
- The workshop should review past decisions made for various species and summarize best practices, which could greatly simplify the content needed within stock assessment reports (e.g., focus text on details specific to the species being assessed)

- **Recommendation for Expanding Reef Fish Observer Program Coverage to the South Atlantic**

- Programmatic funding should be allocated to expand existing observer temporal and spatial coverage in the South Atlantic reef fish fishery. Observer coverage should be sufficient to provide for statistically rigorous discard estimation methods and to provide adequate discard size composition data for use in stock assessments.

RECREATIONAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

- Increase sampling of the recreational fishing fleet, particularly the charter boat and private angler sector, to improve discard data collection. Discard length data and discard mortality are two areas of importance that should be included.
- Continue to develop methods to provide uncertainty estimates around landings and discard estimates
- Investigate the implications of the MRIP imputed lengths and weighting factors for a range of data-rich to data-limited species, where the length frequency distributions become erratic

MEASURES OF POPULATION ABUNDANCE RESEARCH RECOMMENDATIONS

No recommendations were provided.

ASSESSMENT PROCESS RESEARCH RECOMMENDATIONS

- Develop methods to characterize length and age composition of scamp observed on videos from the SERFS fishery-independent survey.
- Implement a systematic age sampling program for both the recreational and commercial sectors.
- Better characterize reproductive parameters including age at maturity, batch fecundity, spawning seasonality, and spawning frequency. Mature male and female biomass was the measure of reproductive potential for scamp in the assessment, but may be biased if reproductive parameters vary significantly with size or age.
- Age-dependent natural mortality was estimated by indirect methods for this assessment of scamp. Mark-recapture approaches (conventional, telemetry, or close-kin) might make it possible to obtain direct estimates of natural mortality of scamp.
- Better characterize the movement dynamics of the stock and the potential for distribution shifts.

REVIEW PANEL RESEARCH RECOMMENDATIONS

The Review Panel supports the research recommendations identified by the Data and Assessment stages for the Gulf of Mexico and South Atlantic assessment processes.

In particular, the Review Panel supported:

- The recommendations to develop artificial intelligence approaches as well as additional automation for image processing and for reading and analysis of video, otoliths, gonad sections and other samples that contribute to scamp stock assessments.

The Review Panel further recommended the following short-term and long-term research needs.

(Short-term, within 6 months)

- Fleet-specific plots of the spatial distribution of the fisheries in both the Gulf and S. Atlantic could help interpret changes in length and age composition over time.
- Dockside sampling was not always randomly structured and in the past, sampling was opportunistic. Investigate modeling issues that may have occurred as a consequence of this.
- For the Gulf, investigate the apparent conflict between the von Bertalanffy model parameters estimated by the model and those provided by the Life History Working Group.
- Further investigation of size and age composition data in the South Atlantic is desirable. Consider “borrowing” length and age composition samples from the Gulf to address poorly sampled strata in the South Atlantic. This assumes that during the historical period, fishery regulations by fleet may have been comparable between the two management units.

(Longer-term)

- Conclude investigation of the taxonomic status of yellowmouth grouper. It has been deemed historically to be a separate species. There is a need to develop a time series of the proportion of yellowmouth grouper over time, perhaps by sampling the catch in the fishery independent series (chevron traps).
- Further investigate changes in reporting of recreational landings from all data sources and how the changes contribute to imprecision in the series.
- Consider the possibility that the ROV data collected by Lewis et al. (2020) could provide another fishery-independent abundance index in the Gulf (see [SEDAR 68 RD44: Changes in Reef Fish Community Structure Following the Deepwater Horizon Oil Spill](#) for a copy of their paper).
- More age samples required for all fleets.
- More effort should be given to formally evaluate and incorporate ecosystem considerations.
- Hold a Best Practices Workshop to address how best to use weights or numbers for recreational harvest in assessment models. This is a much more complex issue than can be resolved in an assessment process.

- **If applicable, provide recommendations for improvement or for addressing any inadequacies identified in the data or assessment modeling. These recommendations should be described in sufficient detail for application, and should be practical for short-term implementation (e.g., achievable within ~6 months). Longer-term recommendations should instead be listed as research recommendations above.**

Additionally, the Review Panel recommends the following:

- The assessment reports could be strengthened by the inclusion of descriptions of the biology and the fishery that are important for the assessment, including information on how management of other species may have affected the fishery in question. For example, in the current case, it was not clear until a late stage of the document that scamp are not directly targeted in the fisheries.
- Move towards a model ensemble of different plausible configurations selected by hypothesis testing and weighed by a comprehensive diagnostic against performance criteria agreed beforehand which is developed to provide stocks status and management advice for both stocks. As best practice, and as a minimum, the ensemble should integrate the three main sources of uncertainty (process uncertainty, parameter uncertainty, and observation error) in the data.
- In these assessments *a priori* assumptions were made about the shape of the selection curve which, while reasonable, there does not seem to be any direct evidence for these fleets that the shape chosen is the right one.
- Currently the Beaufort Assessment Model does not support an option to model discards with a retention function and appears to require this catch category to be modelled as a separate fleet. This does not reflect the way the observations are collected and the model needs to be enhanced to allow discards to be modelled with a separate retention function for the fleet concerned. In addition, having the option in the Beaufort Assessment Model to model selectivity by length would be desirable in the future.
- In order to overcome the problem of changes of scale seen in the Gulf retrospectives a more robust way of expressing F and biomass over time would be to use ratio estimators such as B/B_{MSY} and F/F_{MSY} .

5. Provide recommendations on possible ways to improve the Research Track Assessment process.

- Recognizing that the Research Track process is new, further background on regarding how it differs from other past and present SEDAR assessments would have been helpful.
- Having the involvement of the Chair of Data Working Group could increase the efficiency of this stage of the review.
- We appreciate the inclusion of some ecosystem considerations in the Gulf assessment where red tide and the 2010 Deep Water Horizon oil spill could have important consequences for fisheries; however more effort should be given to formally evaluate and incorporate ecosystem considerations throughout the assessment process.
- Make assessment data and models fully available to panelist. Removing certain data due to confidentiality hinder the work of the reviewers and negatively affects its quality.